

# Sertifikat Kalibrasi

## Calibration Certificate

Certificate number: 9

Order number: 9

### Deskripsi Objek yang Dikalibrasi/Diukur

*Description of object being calibrated or measured*

Jenis alat atau objek : SPRT Pt-25

*Type of instrument or object*

Merek/pembuat dan tipe : Fluke 5681

*Brand/manufacturer and type*

Identifikasi alat

*Instrument identification*

Nomor seri : -

*Serial number*

Identifikasi lain : -

*Other identification*

### Identitas Pemilik

*Owner's identification*

Nama : .

*Designation*

Alamat : . . . , . . . ,

*Address*

### Pengesahan

*Authorization*

Pejabat yang mengesahkan : Direktur SNSU Termoelektrik dan Kimia

*Authorizing officer*

*(Director of National Measurement Standards for Thermoelectricity and Chemistry)*

Nama : Dr. Ghufon Zaid

*Name*

NIP 19711104 199012 1 001

Tanggal pengesahan :

*Date of issue (dd/mm/yyyy)*

Jumlah halaman (termasuk : 3

halaman ini)

*Total number of pages including this one*



Dokumen ini disahkan secara elektronik sesuai peraturan yang berlaku dengan sertifikat dari Balai Sertifikasi Elektronik (BsrE) dan tidak memerlukan tanda tangan atau cap. Dokumen asli dapat diperoleh dengan memindai kode QR di samping ini.

*This document is digitally signed. No signature or seal is required. The original document can be obtained by scanning the QR code on the left.*

Kalibrasi atau pengukuran yang dilaporkan dalam sertifikat ini tercakup dalam lingkup akreditasi menurut SNI ISO/IEC 17025 oleh Komite Akreditasi Nasional, kecuali dinyatakan dalam badan sertifikat.

*The calibration or measurement reported in the certificate is covered in the accreditation scope according to SNI ISO/IEC 17025 by the National Accreditation Committee of Indonesia, unless marked otherwise in the body of certificate.*

Nama Alat/*Instrument Name* : SPRT Pt-25  
Pembuat/*Manufacturer* : Fluke  
Model/*Model* : 5681  
No. Seri/*Serial Number* : -  
Tanggal Kalibrasi/*Calibration Date* : 24 Oktober 2025 - 25 Oktober 2025  
Tempat Kalibrasi/*Calibration Place* : Laboratorium SNSU-BSN

### Hasil Kalibrasi/*Calibration Result*

#### **Kondisi Ruangan/*Environmental Condition***

Suhu/*Temperature* :  $(21.0 \pm 0.0) ^\circ\text{C}$   
Kelembapan/*Humidity* :  $(62.0 \pm 0.0) \%$

#### **Hasil Kalibrasi SSPRT Pt-100 Menggunakan Metode Titik Tetap / *Calibration Results of SSPRT Pt-100 using Fixed Points Method***

| <b>Definisi Suhu</b><br><i>Temp. Definitions</i> | <b>Penunjukkan SPRT</b><br><i>SPRT Indications</i> | <b>Ketidakpastian</b><br><i>Uncertainty</i> |
|--------------------------------------------------|----------------------------------------------------|---------------------------------------------|
| -38.8344<br>element: list<br>object[0] }}        | 21.492790<br>element: list<br>object[0] }}         | 4.3<br>element: list<br>object[0] }}        |
| 0.01<br>element: list<br>object[1] }}            | 25.459708<br>element: list<br>object[1] }}         | 1.2<br>element: list<br>object[1] }}        |
| 29.7646<br>element: list<br>object[2] }}         | 28.466701<br>element: list<br>object[2] }}         | 5.1<br>element: list<br>object[2] }}        |
| 231.928<br>element: list<br>object[3] }}         | 48.183526<br>element: list<br>object[3] }}         | 8.0<br>element: list<br>object[3] }}        |
| 419.527<br>element: list<br>object[4] }}         | 65.392049<br>element: list<br>object[4] }}         | 10.0<br>element: list<br>object[4] }}       |
| 660.323<br>element: list<br>object[5] }}         | 85.933750<br>element: list<br>object[5] }}         | 10.0<br>element: list<br>object[5] }}       |

#### **Catatan/*Notes***

Termometer Tahanan Platina Semi Standar (TTPS) Pt-100 dengan spesifikasi di atas dikalibrasi dengan metode titik tetap berdasarkan skala suhu internasional tahun 1990 (ITS-90) pada sub-rentang W<sub>5</sub> (-38,8344 °C — 29,7646 °C) dan sub-rentang W<sub>7</sub> (0.01 °C — 660.323 °C). /

*The Semi Standard Platinum Resistance Thermometer (SPRT) Pt-100 with above specifications is calibrated by fixed point method in accordance with the International Temperature Scale of 1990 (ITS-90) in the sub-range W<sub>5</sub> (-38,8344 °C — 29,7646 °C) and in the sub-range W<sub>7</sub> (0.01 °C — 660.323 °C).*

Konversi suhu dalam derajat celcius (t<sub>90</sub>/°C) dapat diperoleh dengan persamaan ITS-90 sebagai berikut. /

*The temperature conversion in unit of degree celcius (t<sub>90</sub>/°C) can be obtained using the following ITS-90 equations.*

Koefisien-koefisien D<sub>i</sub> dan B<sub>i</sub> diperoleh dari dokumen ITS-90 sebagai berikut: D<sub>0</sub> = 439,932 854 ; D<sub>1</sub> = 472,418 020 ; D<sub>2</sub> = 37,684 494 ; D<sub>3</sub> = 7,472 018 ; D<sub>4</sub> = 2,920 828 ; D<sub>5</sub> = 0,005 184 ; D<sub>6</sub> = -0,963 864 ; D<sub>7</sub> = -0,188 732 ; D<sub>8</sub> = 0,191 203 ; D<sub>9</sub> = 0,049 025 ; B<sub>0</sub> = 0,183 324 722 ; B<sub>1</sub> = 0,240 975 303 ; B<sub>2</sub> = 0,209 108 771 ; B<sub>3</sub> = 0,190 439 972 ; B<sub>4</sub> = 0,142 648 498 ; B<sub>5</sub> = 0,077 993 465 ; B<sub>6</sub> = 0,012 475 611 ; B<sub>7</sub> = -0,032 267 127 ; B<sub>8</sub> = -0,075 291 522 ; B<sub>9</sub> = -0,056 470 670 ; B<sub>10</sub> = 0,076 201 285 ; B<sub>11</sub> = 0,123 893 204 ; B<sub>12</sub> = -0,029 201 193 ; B<sub>13</sub> = -0,091 173 542 ; B<sub>14</sub> = 0,001 317 696 ; B<sub>15</sub> = 0,026 025 526 /

The coefficients  $D_i$  and  $B_i$  can be obtained from the ITS-90 official document as follows:  $D_0 = 439,932\,854$ ;  $D_1 = 472,418\,020$ ;  $D_2 = 37,684\,494$ ;  $D_3 = 7,472\,018$ ;  $D_4 = 2,920\,828$ ;  $D_5 = 0,005\,184$ ;  $D_6 = -0,963\,864$ ;  $D_7 = -0,188\,732$ ;  $D_8 = 0,191\,203$ ;  $D_9 = 0,049\,025$ ;  $B_0 = 0,183\,324\,722$ ;  $B_1 = 0,240\,975\,303$ ;  $B_2 = 0,209\,108\,771$ ;  $B_3 = 0,190\,439\,972$ ;  $B_4 = 0,142\,648\,498$ ;  $B_5 = 0,077\,993\,465$ ;  $B_6 = 0,012\,475\,611$ ;  $B_7 = -0,032\,267\,127$ ;  $B_8 = -0,075\,291\,522$ ;  $B_9 = -0,056\,470\,670$ ;  $B_{10} = 0,076\,201\,285$ ;  $B_{11} = 0,123\,893\,204$ ;  $B_{12} = -0,029\,201\,193$ ;  $B_{13} = -0,091\,173\,542$ ;  $B_{14} = 0,001\,317\,696$ ;  $B_{15} = 0,026\,025\,526$

$W_r(t_{90})$  dapat diperoleh dari persamaan berikut: /

$W_r(t_{90})$  can be obtained using the following equations:

$W(t_{90})$  dapat diperoleh dari persamaan berikut, di mana  $R(t_{90})$  adalah tahanan SSPRT pada suhu  $t$  dalam satuan  $\Omega$  dan  $R(t_{TPW})$  adalah tahanan SSPRT pada titik tripel air ( $0,01\,^{\circ}\text{C}$ ) dalam satuan  $\Omega$ . /

$W(t_{90})$  can be obtained using the following equations, where  $R(t_{90})$  is the SPRT resistance at a temperature  $t$  in unit  $\Omega$  and  $R(t_{TPW})$  is the SPRT resistance at triple point of water ( $0,01\,^{\circ}\text{C}$ ) in unit  $\Omega$ .

Koefisien-koefisien  $a_5$ ,  $b_5$ ,  $a_7$ ,  $b_7$ , dan  $c_7$  diperoleh dari hasil kalibrasi pada arus eksitasi 1 mA:  $a_5 = -2,930\,475\,\text{E-04}$ ;  $b_5 = 2,563\,645\,\text{E-04}$ ;  $a_7 = -2,717\,029\,\text{E-04}$ ;  $b_7 = -1,586\,798\,\text{E-05}$ ;  $c_7 = 1,043\,757\,\text{E-06}$  /

The coefficients  $a_5$ ,  $b_5$ ,  $a_7$ ,  $b_7$ , and  $c_7$  were obtained from the calibration at excitation current of 1 mA:  $a_5 = -2,930\,475\,\text{E-04}$ ;  $b_5 = 2,563\,645\,\text{E-04}$ ;  $a_7 = -2,717\,029\,\text{E-04}$ ;  $b_7 = -1,586\,798\,\text{E-05}$ ;  $c_7 = 1,043\,757\,\text{E-06}$

Ketidakpastian rambatan dalam satuan mK sebagai fungsi suhu  $U(t_{90})$  pada rentang-rentang kalibrasi tersebut dapat diperkirakan dengan grafik berikut: /

The propagated uncertainty in unit of mK, as the function of temperature  $U(t_{90})$  in the mentioned calibration ranges, can be estimated using the following chart:

Semua nilai ketidakpastian pada pengukuran ini dinyatakan pada tingkat kepercayaan 95% dengan faktor cakupan  $k = 2$  /

All the uncertainty values in this measurement are expressed at 95% confidence level with coverage factor of  $k = 2$

Alat standar yang digunakan adalah Hg TP Cell Fluke 5900 (SN. Hg 00001), H<sub>2</sub>O TP Cell PTB - (SN. PTB4), Ga MP Cell Fluke HS 5943 (SN. 43013), Sn FP Cell Fluke HS 5925 (SN. 05065), Zn FP Cell Fluke HS 5926 (SN. 06070), Al FP Cell Fluke HS 5927 (SN. 07096), dan Thermometry Bridge Isotech MicroK 70 20-P2273 (SN. ITL42569-1). /

The standard instruments used were Hg TP Cell Fluke 5900 (SN. Hg 00001), H<sub>2</sub>O TP Cell PTB - (SN. PTB4), Ga MP Cell Fluke HS 5943 (SN. 43013), Sn FP Cell Fluke HS 5925 (SN. 05065), Zn FP Cell Fluke HS 5926 (SN. 06070), Al FP Cell Fluke HS 5927 (SN. 07096), and Thermometry Bridge Isotech MicroK 70 20-P2273 (SN. ITL42569-1).

Kalibrasi ini dilakukan pada rentang pengukuran  $(-38,8344 \sim 660,323)\,^{\circ}\text{C}$  /

The calibration is performed within the measurement range  $(-38,8344 \sim 660,323)\,^{\circ}\text{C}$

Hasil kalibrasi tersebut tertelusur ke satuan pengukuran SI melalui Laboratorium Standar Nasional Satuan Ukuran (SNSU) — Badan Standardisasi Nasional (BSN). /

The calibration results are traceable to the SI unit through the Laboratory of National Measurement Standards (NMS) — The National Standardization Agency (BSN).

Dikalibrasi oleh/Calibrated by : Dwi Larassati, S.T. & Kelvin Sapta Dewantara, S.Si.

Diperiksa oleh/Checked by : Dr. Aditya Achmadi, S.Si., M.T.

(Penyelia/Supervisor)

: Dr. Aditya Achmadi, S.Si., M.T.

(Ka. Lab SNSU Suhu/Head of NMS for Temperature Laboratory)

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===== Akhir dari Sertifikat/End of Certificate =====